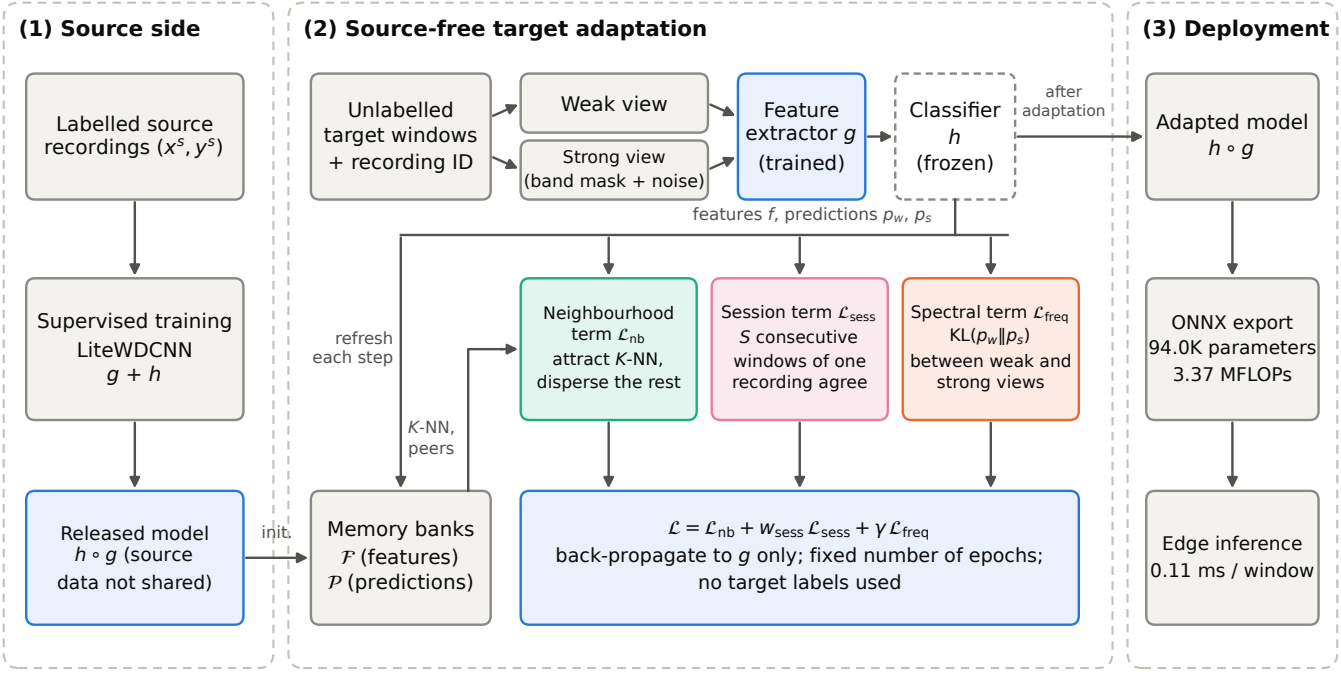
(1) Source side
Labelled source recordings (x^s, y^s)
Supervised training LiteWDCNN $g + h$
Released model $h \circ g$ (source data not shared)
init.
(2) Source-free target adaptation
Unlabelled target windows + recording ID
Weak view
Strong view (band mask + noise)
Feature extractor g (trained)
Classifier h (frozen)
after adaptation
features f, predictions p_w, p_s
refresh each step
K-NN, peers
Neighbourhood term $\mathcal{L}_{nb}$ attract K-NN, disperse the rest
Session term $\mathcal{L}_{sess}$ S consecutive windows of one recording agree
Spectral term $\mathcal{L}_{freq}$ KL($p_w \| p_s$) between weak and strong views
Memory banks $\mathcal{F}$ (features) $\mathcal{P}$ (predictions)
$\mathcal{L} = \mathcal{L}_{nb} + w_{sess}\mathcal{L}_{sess} + \gamma\mathcal{L}_{freq}$
back-propagate to g only; fixed number of epochs; no target labels used
(3) Deployment
Adapted model $h \circ g$
ONNX export 94.0K parameters 3.37 MFLOPs
Edge inference 0.11 ms / window